Fundamentals 29

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Problem 29.1

Part (1)

The subset binary operation always forms a poset on any set.

Part (2)

Let B be a maximal element of $\mathcal{P}$. So for all $B' \in \mathcal{P}$, $B \supsetneq B'$. The largest possible size of B' is 3, so B has 3 linearly independent elements. Therefore, it is a basis.

Part (3)

Assume $\mathcal{P}$ contains an upper bound B. Then for all $B' \in \mathcal{P}$, $B' \subseteq B$. But we can always choose a different linearly independent subset that doesn't share elements with B.

Problem 29.2

Part (1)

No, it doesn't satisfy property (2). If $N(z) \leq N(y)$ and $N(y) \leq N(z)$ then z and y have the same magnitude but they aren't necessarily the same.

Part (2)

It has no upper bound or maximal element.

Part (3)

Yes, because:

1. For all $x \in \mathbb{C}$, $z \not< z$.
2. If $N(x) < N(y)$ and $N(y) < N(z)$, then $N(x) < N(z)$ so $x < z$.
3. For all $x, y \in \mathbb{C}$, either $N(x) < N(y)$, $N(y) > N(x)$, or $N(x) = N(y)$, so either $x < y$, $y > x$, or $x = y$.

Problem 29.3

Let $\{X_i : i \in \mathbb{N}\}$ be a countable set where each X_i is countable, and let $X = \bigcup_{i\in\mathbb{N}} X_i$. Since each X_i is countable, there are bijections $f_i : \mathbb{N} \to X_i$. Define $f : \mathbb{N} \to X$ as

$$\begin{aligned} f(1) &= f_1(1), \\ f(2) &= f_2(1), \\ f(3) &= f_1(2), \\ f(4) &= f_1(3), \\ f(5) &= f_2(2), \\ f(6) &= f_3(1), \\ f(7) &= f_4(1), \end{aligned}$$

and so on in a diagonal pattern. Then f is a bijection because if $x \in X$ then $x \in X_i$ for some i, so $x = f_i(j)$ for some j.

The fact that we can find bijections f_i for all $i \in \mathbb{N}$ is implied by AC.

Problem 29.5

Let

$$\mathcal{P} = \{\mathcal{G} \subseteq \mathcal{P}(X) : \mathcal{G} \text{ is a filter on } X \text{ and } \mathcal{F} \subseteq \mathcal{G}\}.$$

Then $(\mathcal{P}, \subseteq)$ is a poset. Let C be a chain. Define

$$\mathcal{G}' = \bigcup_{\mathcal{G}\in C} \mathcal{G}.$$

We claim that $\mathcal{G}'$ itself is a filter on X containing $\mathcal{F}$, which therefore lies in $\mathcal{P}$ and is an upper bound for C.

1. Since each $\mathcal{G} \in C$ is a proper filter, $\varnothing \notin \mathcal{G}'$.
2. If $A, B \in \mathcal{G}'$, there exist $\mathcal{G}_1, \mathcal{G}_2 \in C$ with $A \in G_1$ and $B \in \mathcal{G}_2$. Since C is a chain, one of $\mathcal{G}_1$, $\mathcal{G}_2$ is a superset of the other, say $\mathcal{G}_1$. Then $A, B \in \mathcal{G}_1$. Since $\mathcal{G}_1$ is a filter, $A \cap B \in \mathcal{G}_1 \subseteq \mathcal{G}'$.
3. If $A \in \mathcal{G}'$ and $A \subseteq B \subseteq X$ then $A \in \mathcal{G}$ for some $\mathcal{G} \in C$. Since $\mathcal{G}$ is a filter, $B \in \mathcal{G} \subseteq \mathcal{G}'$.

By Zorn's Lemma, $\mathcal{P}$ has a maximal element, $\mathcal{G} \in \mathcal{P}$. Since the finest elements of filters are ultrafilters, $\mathcal{G}$ is an ultrafilter on X.

Problem 29.7

1. We know that $(a, b) \leq (a, b)$ since $a \leq a$ and $b \leq b$.

2. Let $(a, b) \leq (c, d)$ and $(c, d) \leq (a, b)$. Then $a \leq c$, $b \leq d$, $c \leq a$, $d \leq b$, so $(a, b) = (c, d)$.

3. Let $(a, b) \leq (c, d)$ and $(c, d) \leq (e, f)$. By transitivity of $\leq$ on $\mathbb{R}$, $(a, b) \leq (e, f)$.

This poset doesn't create a linear order, since there are points in $\mathbb{R}^2$ that aren't comparable. For example, none of the following are true: $(0, 1) \geq (1, 0)$, $(1, 0) \leq (0, 1)$, $(1, 0) = (0, 1)$.

Problem 29.8

This is not a poset because antisymmetry fails. We have that $(0, 1) \leq (1, 0)$ and $(1, 0) \leq (0, 1)$ but $(0, 1) \neq (1, 0)$. This also doesn't create a linear ordering, since if we let $x = (2, 0)$, $y = (0, 1)$, $z = (1, -1)$, then $x < y$ and $y < z$ but $x \not< z$.